

Project: Programming Language C++
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Target audience: SG1, LWG

P0523r0: Wording for CH 10: Complexity of parallel algorithms

Proposed wording

In 25.2.5 [algorithms.parallel.overloads] add another paragraph after p2:

Unless otherwise specified, the complexity requirements of `ExecutionPolicy` algorithm overloads are relaxed from the complexity requirements of the overloads without as follows:

When the guarantee says "At most *expr*" or "Exactly *expr*" and doesn't specify the number of assignments or swaps, and *expr* isn't already an O-Notation, the complexity of the algorithm shall be " $O(expr)$ ".

In 25.4.4p4 [alg.transform], change:

Complexity: Exactly $last1 - first1$ applications of `op` or `binary_op`. This requirement also applies to the overload with an `ExecutionPolicy`.

In 25.5.4p8 [alg.merge], change:

Complexity: When enough additional memory is available, $(last - first) - 1$ comparisons. For the overloads with an `ExecutionPolicy` the complexity is $O(last - first)$. If no additional memory is available, an algorithm with complexity $N \log(N)$ may be used, where $N = last - first$.

Acknowledgments

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